Cloud Computing: Deploying a Static Website on AWS S3

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1. Introduction

Cloud computing means using servers, storage, databases and software over the internet instead of owning and maintaining them. Users pay only for what they use and can scale up or down quickly. In this mission, I learned the main cloud service models, created an AWS Free Tier account, and deployed a simple static website on Amazon S3.

2. IaaS vs PaaS vs SaaS

The three service models differ in how much the provider manages and how much the user manages.

- IaaS (Infrastructure as a Service) :The provider gives raw computing resources such as virtual machines, storage and networks. The user installs and manages the OS, software and security. Example:Amazon EC2, Google Compute Engine, Azure Virtual Machines
- PaaS (Platform as a Service) :The provider manages the infrastructure, OS and runtime. The user only deploys code and manages the application. Example:Heroku, Google App Engine, Vercel
- SaaS (Software as a Service) :A finished application used over the internet. The provider manages everything. Example:Gmail, Netflix, Zoom, Dropbox

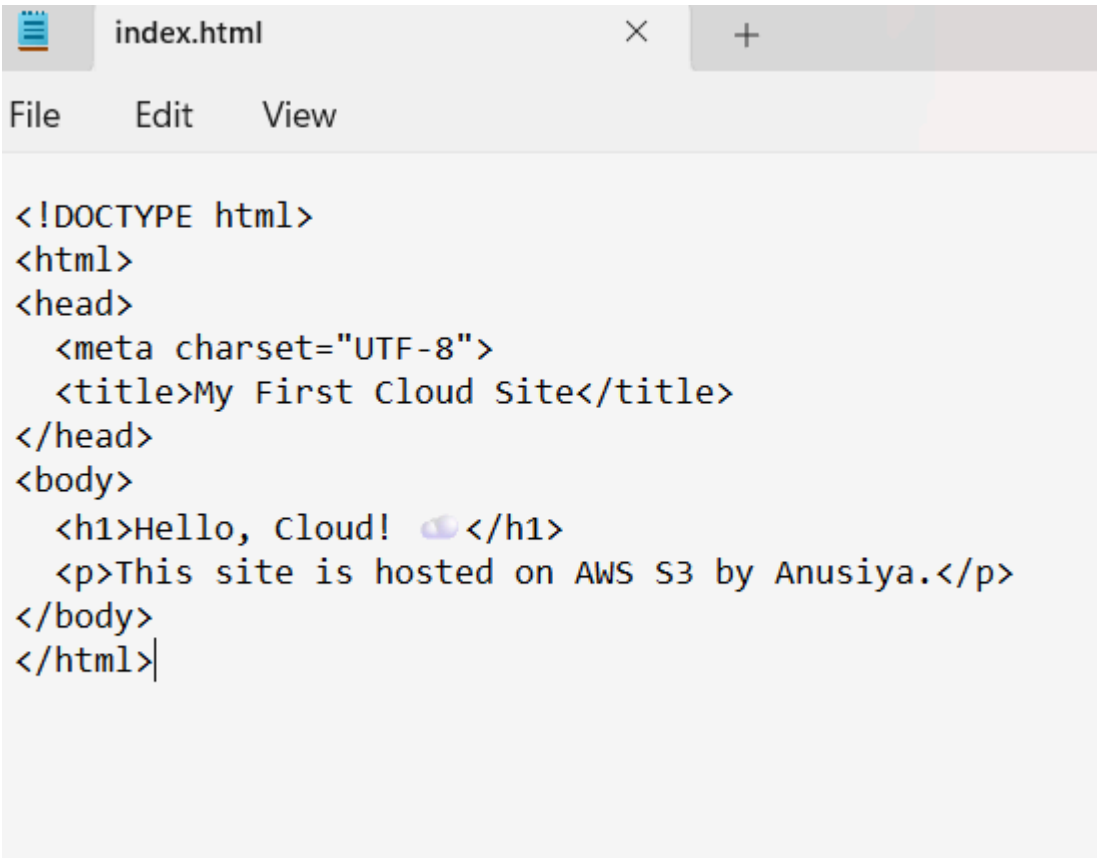
In short, the user does the most work in IaaS, some work in PaaS, and the least work in SaaS. Amazon S3, which I used in this mission, is a storage service that falls under the IaaS category.

3. What I Did

Created an AWS Free Tier account. I signed up at aws.amazon.com, completed identity verification, added a payment method for verification, and chose the Basic (free) support plan.

Created an S3 bucket. In the S3 service, I created a general purpose bucket named `anusiya-my-first-site-579096219414-ap-southeast-2-an` in the Asia

Pacific (Sydney) region and turned off Block Public Access so the site could be viewed publicly.
Uploaded the website file. I wrote a simple `index.html`



Enabled static website hosting. In the bucket Properties tab, I enabled Static website hosting and set `index.html` as the index document.

Added a bucket policy. In the Permissions tab, I added a bucket policy that allows public read access (`s3:GetObject`) to the objects in the bucket.

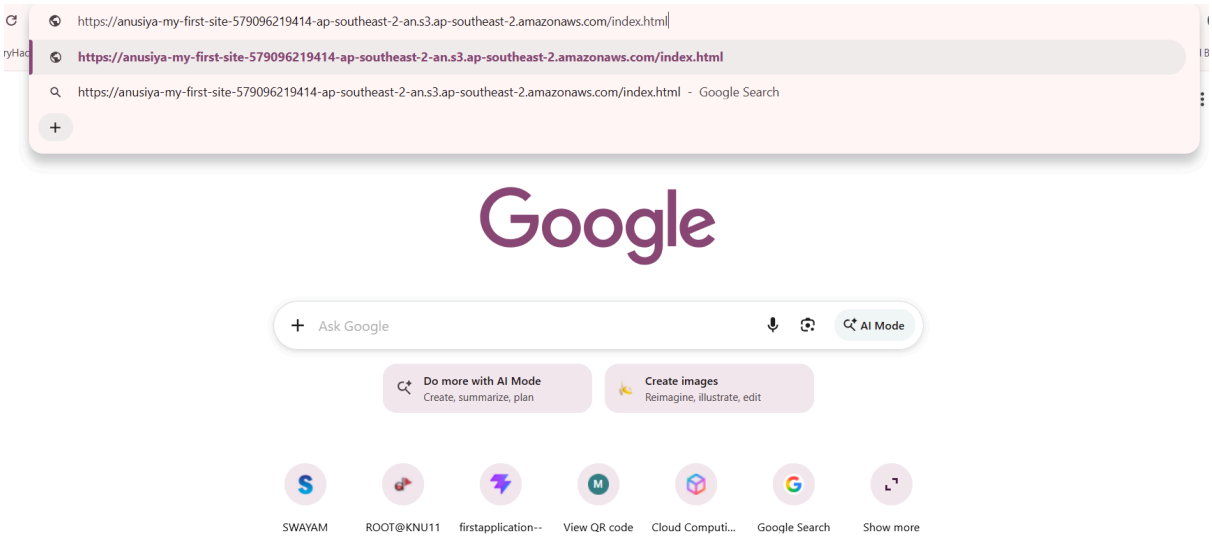
The website. I opened the bucket website endpoint in a browser and the page loaded successfully.

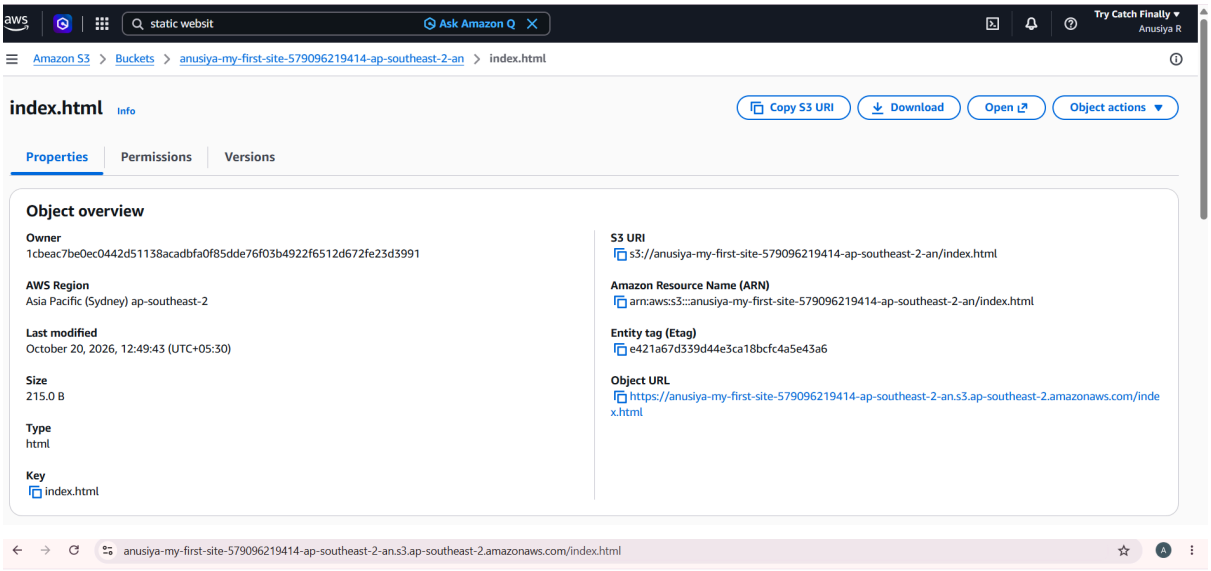
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4. Cloud Architecture

The diagram below shows how a visitor reaches the website.

You (Browser) → Internet → Amazon S3 bucket → index.html







Hello, Cloud!

This site is hosted on AWS S3 by Anusiya.

When a user opens the website URL, the browser sends a request over the internet to Amazon S3. S3 finds `index.html` in the bucket and sends it back, and the browser displays it. There is no server to manage, because S3 serves the file directly.

5. Challenges and Learnings

Challenge: After deployment, the emoji on my page appeared as broken characters instead of the cloud symbol. The browser did not know which character encoding the page used.

Solution: I added `<meta charset="UTF-8">` inside the `<head>` section of `index.html`, saved the file with UTF-8 encoding, and uploaded it again. The page then displayed correctly.

Other learnings:

- The bucket name must be globally unique, and AWS may add extra text to it.
- Block Public Access must be turned off before a public bucket policy can be saved.
- Public buckets should never hold personal or sensitive data.

6. Conclusion

This mission helped me understand the difference between IaaS, PaaS and SaaS and gave me hands-on experience with AWS. I learned how to create a cloud account, use Amazon S3, and host a website without managing any servers. I also learned the importance of security settings such as public access and the need to delete unused resources to avoid charges.

7. Deployment URL

LINK :
http://anusiya-my-first-site-579096219414-ap-southeast-2-an.s3-website-ap-southeast-2.amazonaws.com